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TWO-WAY ANGLE VALVE CAPABLE OF BUFFERING DEFLATION

Abstract

A two-way angle valve capable of buffering deflation includes: a body having an inlet and an outlet; a drive cylinder coupled to the body; a main piston having therein an auxiliary drive cavity; a drive air duct; an auxiliary piston driven to be vertically movably disposed in the auxiliary drive cavity; an auxiliary spring; a stem fixedly disposed at the auxiliary piston and having an auxiliary disc; a main disc having a chamber, deflation buffering duct and deflation buffering hole; and a main spring. The stem is inserted into the main disc such that the auxiliary disc is vertically movably disposed in the chamber. The deflation buffering duct has one end in communication with the chamber and the other end defined as an opening formed at the main disc laterally. Therefore the deflation buffering duct and deflation buffering hole are in communication with the outlet and inlet respectively.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present disclosure relates to angle valves, and more particularly to a two-way angle valve capable of buffering deflation.

2. Description of Related Art

[0002] Taiwan patent 1551797 discloses a two-way angle valve with an airtight disc and a first channel. The airtight disc undergoes stroke control to reach a first distance and a second distance relative to the first channel, thereby controlling gas flow rate by ensuring that the first distance is less than the second distance. However, Taiwan patent 1551797 has a drawback. The aforesaid controlling the gas flow rate by controlling the extent of the stroke of the airtight disc does not make a significant difference in the gas flow rate between the first distance and the second distance for the reason explained below. Different strokes but the same area of the open airtight disc exist at the first distance and the second distance. The gas flow rate is already high during the time period from the point in time when the airtight disc is opened to the point in time of the first distance, and in consequence the difference in the gas flow rate will not vary significantly even if the airtight disc is opened further to reach the second distance.

[0003] China patent 116817003A discloses a two-way angle valve that improves the prior art and comprises a main disc and an auxiliary disc. The main disc and the auxiliary disc open separately to controllably cause a main inlet and an auxiliary inlet to open or shut. The auxiliary inlet has a smaller diameter than the main inlet. The auxiliary inlet is opened to let a small amount of gas in, and then the main inlet is opened to let a large amount of gas in, buffering gas admission. However, China patent 116817003A has a drawback. The two-way angle valve has a main stem and an auxiliary stem which have to be driven by a drive mechanism each. Specifically speaking, the main stem is driven by a main piston, and the auxiliary stem is driven by an auxiliary piston; doing so, however, requires fitting the auxiliary stem inside the main stem, requires keeping their movement smooth and independent of each other, and requires maintaining airtightness with an O-ring. As a result, the two-way angle valve is structurally complicated and difficult to assemble, not to mention that some components of the two-way angle valve fail to operate smoothly. Furthermore, gas may seep into the auxiliary disc and come into contact with an auxiliary resilient storing element. If the gas is corrosive, the auxiliary resilient storing element may be corroded, causing a failure of the two-way angle valve.

BRIEF SUMMARY OF THE INVENTION

[0004] It is an objective of the disclosure to provide a two-way angle valve capable of buffering deflation.

[0005] Another objective of the disclosure is to provide a two-way angle valve capable of buffering deflation, using a stem to drive an auxiliary disc and a main disc separately to not only achieve structural simplicity and ease of assembly but also ensure that all the components of the two-way angle valve operate smoothly.

[0006] To achieve the above and other objectives, the disclosure provides a two-way angle valve capable of buffering deflation, comprising: a body having an inlet disposed below and an outlet disposed laterally; a drive cylinder coupled to the body and disposed on the body; a main piston driven to be vertically movably disposed in the drive cylinder, wherein an auxiliary drive cavity is

concavely disposed at the main piston and extends in a top-to-bottom direction; a drive air duct not only disposed at the drive cylinder so as to be in communication with the inside of the drive cylinder but also externally connected to a controlled gas source; an auxiliary piston driven to be vertically movably disposed in the auxiliary drive cavity; an auxiliary spring disposed in the drive cylinder, having two ends abutting against the drive cylinder and the auxiliary piston respectively, and generating a resilient restoring force under which the auxiliary piston is moved downward; a stem having a top end fixedly disposed at the auxiliary piston, extending downward by a predetermined length, and moving vertically together with the auxiliary piston to penetrate the main piston and the drive cylinder to thereby move into the body, wherein an auxiliary disc is disposed at the bottom end of the stem; an auxiliary drive air duct comprising an auxiliary channel penetrating the drive cylinder, a gap defined between the stem and the drive cylinder, and a stem air duct disposed in the stem, wherein the auxiliary channel, the gap, and the stem air duct are in communication with each other, the stem air duct being in communication with the inside of the auxiliary drive cavity, and the auxiliary channel being externally connected to an auxiliary controlled gas source; a main disc having a chamber and a deflation buffering duct, the main disc having an upper abutting portion and a lower abutting portion, the upper abutting portion being disposed above the chamber, the lower abutting portion being disposed below the chamber and having a deflation buffering hole, the stem being inserted into the main disc to allow the auxiliary disc to be vertically movably disposed in the chamber, the deflation buffering duct having one end being in communication with the chamber and the other end being defined as an opening formed on a lateral side of the main disc, thereby allowing the deflation buffering duct to be in communication with the outlet, and the deflation buffering hole being in communication with the inlet; and a main spring disposed in the body, fitted around the stem, having two ends abutting against the drive cylinder and the main disc respectively, and generating a resilient restoring force under which the main disc is moved downward, wherein the stem and the auxiliary disc move because of vertical movement of the auxiliary piston, wherein movement of the auxiliary piston and the main piston causes the stem to move between a lowest position and a highest position, wherein when the stem is at the lowest position, the auxiliary disc abuts against the lower abutting portion of the chamber to close the deflation buffering hole, and the main disc abuts against the body to close the inlet, disabling communication of the inlet and the outlet, wherein when the auxiliary disc has separated from the lower abutting portion but has not yet abutted against the upper abutting portion during an ascent of the stem from the lowest position, the outlet is in communication with the inlet through the deflation buffering duct, the chamber and the deflation buffering hole, wherein when the stem is at the highest position, the auxiliary disc abuts against the upper abutting portion of the chamber, and the main disc is lifted such that the inlet is not closed, enabling the communication of the inlet and the outlet.

[0007] Owing to the aforesaid technical features, the two-way angle valve of the disclosure not only achieves deflation buffering but also uses a stem to drive an auxiliary disc and a main disc separately to achieve structural simplicity and ease of assembly and ensure that all the components of the two-way angle valve operate smoothly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a preferred embodiment of the disclosure.

[0009] FIG. 2 is a cross-sectional view taken along line 2-2 of FIG. 1.

[0010] FIG. 3 is a cross-sectional view taken along line 3-3 of FIG. 1.

[0011] FIG. 4 is a partial enlarged view of FIG. 3.

[0012] FIG. 5 is a cross-sectional view of operation of a preferred embodiment of the disclosure.

[0013] FIG. 6 is a partial enlarged view of FIG. 5.

[0014] FIG. 7 is another cross-sectional view of operation of a preferred embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0015] Technical features of the disclosure are herein illustrated with preferred embodiments, depicted with drawings, and described below.

[0016] As shown in FIG. 1 through FIG. 7, a preferred embodiment of the disclosure provides a two-way angle valve **10** capable of buffering deflation essentially comprising a body **11**, a drive cylinder **15**, a main piston **21**, a drive air duct **25**, an auxiliary piston **31**, an auxiliary spring **35**, a stem **41**, an auxiliary drive air duct **45**, a main disc **51** and a main spring **55**.

[0017] An inlet **12** is disposed below the body **11**. An outlet **13** is disposed beside the body **11**.

[0018] The drive cylinder **15** is coupled to the body **11** and disposed on the body **11**. In practice, to facilitate processing and production, the drive cylinder **15** comprises a lower block **151** and an upper cylinder **152** fitted together. The body **11** is disposed at the lower block **151**. A drive cavity **153** is formed between the upper cylinder **152** and the lower block **151**.

[0019] The main piston **21** is driven to be vertically movably disposed in the drive cavity **153** of the drive cylinder **15**. An auxiliary drive cavity **213** is concavely disposed at the main piston **21** and extends in a top-to-bottom direction.

[0020] The drive air duct **25** is disposed at the drive cylinder **15** and thus is in communication with the inside of the drive cylinder **15**. The drive air duct **25** is externally connected to a controlled gas source **91**. In practice, the drive air duct **25** is disposed at the lower block **151**.

[0021] The auxiliary piston **31** is driven to be vertically movably disposed in the auxiliary drive cavity **213**.

[0022] The auxiliary spring **35** is disposed in the drive cavity **153** of the drive cylinder **15**. Two ends of the auxiliary spring **35** abut against the upper cylinder **152** and the auxiliary piston **31** respectively. The auxiliary spring **35** generates a resilient restoring force under which the auxiliary piston **31** is moved downward.

[0023] The top end of the stem **41** is fixedly disposed at the auxiliary piston **31**, extends downward by a predetermined length to penetrate the lower block **151**, and moves vertically together with the auxiliary piston **31** to penetrate the main piston **21** and the drive cylinder **15** and thereby move into the body **11**. An auxiliary disc **42** is disposed at the bottom end of the stem **41**.

[0024] The auxiliary drive air duct **45** comprises an auxiliary channel **451** penetrating the drive cylinder **15**, a gap **G** defined between the stem **41** and the drive cylinder **15**, and a stem air duct **411** disposed in the stem **41**. The auxiliary channel **451**, the gap **G**, and the stem air duct **411** are in communication with each other. The stem air duct **411** is in communication with the inside of the auxiliary drive cavity **213**. The auxiliary channel **451** is externally connected to an auxiliary controlled gas source **92**. In this embodiment, the auxiliary channel **451** is disposed at the lower block **151** for exemplary purposes.

[0025] The main disc **51** has a chamber **52** and a deflation buffering duct **53**. The main disc **51** has an upper abutting portion **511** and a lower abutting portion **512** which are disposed above and below the chamber **52** respectively. The lower abutting portion **512** has a deflation buffering hole **513**. The stem **41** is inserted into the main disc **51** to allow the auxiliary disc **42** to be vertically movably disposed in the chamber **52**. The deflation buffering duct **53** has one end being in communication with the chamber **52** and the other end being defined as an opening formed on a lateral side of the main disc **51**. The deflation buffering duct **53** is in communication with the outlet **13**, and the deflation buffering hole **513** is in communication with the inlet **12**. In this embodiment, the diameter of the deflation buffering duct **53** is configured to be less than the diameter of the inlet **12**, and the diameter of the deflation buffering hole **513** is also configured to be less than the diameter of the inlet **12**, ensuring that the deflation buffering duct **53** and the deflation buffering hole **513** are of diameters small enough to admit a working gas at a low flow rate and thereby

buffer the atmospheric pressure difference between the inlet **12** and the outlet **13**.

[0026] The main spring **55** is disposed in the body **11** and fitted around the stem **41**. The main spring **55** has two ends abutting against the drive cylinder **15** and the main disc **51** respectively and generates a resilient restoring force under which the main disc **51** is moved downward. If the two-way angle valve **10** capable of buffering deflation is used in an environment with a working gas being corrosive, this embodiment further provides a resilient sealing pipe **56** with two ends connected to the main disc **51** and the body **11** respectively to cover the stem **41** and the main spring **55**, insulating the main spring **55** from the corrosive working gas.

[0027] The stem **41** and the auxiliary disc **42** move because of the vertical movement of the auxiliary piston **31**. In practice, the lower block **151** has an upper seal ring **S1** and a lower seal ring **S2** and is slidably fitted to the stem **41** so as to allow the position of communication of the auxiliary channel **451** and the gap **G** to stay between the upper seal ring **S1** and the lower seal ring **S2** and allow the position of communication of the stem air duct **411** and the gap **G** to stay between the upper seal ring **S1** and the lower seal ring **S2** regardless of how the stem **41** moves.

[0028] The movement of the auxiliary piston **31** and the main piston **21** causes the stem **41** to move between a lowest position **LP** and a highest position **HP**. When the stem **41** is at the lowest position **LP**, the auxiliary disc **42** abuts against the lower abutting portion **512** of the chamber **52** to close the deflation buffering hole **513**, and the main disc **51** abuts against the body **11** to close the inlet **12**, disabling the communication of the inlet **12** and the outlet **13**. When the auxiliary disc **42** has separated from the lower abutting portion **512** but has not yet abutted against the upper abutting portion **511** during an ascent of the stem **41** from the lowest position **LP**, the outlet **13** is in communication with the inlet **12** through the deflation buffering duct **53**, the chamber **52** and the deflation buffering hole **513**. When the stem **41** is at the highest position **HP**, the auxiliary disc **42** abuts against the upper abutting portion **511** of the chamber **52**, and the main disc **51** is lifted such that the inlet **12** is not closed, enabling the communication of the inlet **12** and the outlet **13**.

[0029] The structural features of this embodiment are described above. The operation of this embodiment is described below. The operation of this embodiment falls into three categories, namely shut, buffered opening and fully opened, and a gas inside the inlet **12** of the body **11** is defined as a working gas.

[0030] As shown in FIG. 2 and FIG. 3, in the shut state, the stem **41** is at the lowest position **LP** to allow the inlet **12** to be closed by the main disc **51** and allow the deflation buffering hole **513** to be closed by the auxiliary disc **42**.

[0031] As shown in FIG. 5, performing buffered opening in the shut state entails the following: a driving gas is provided to the auxiliary drive air duct **45** by the auxiliary controlled gas source **92**, and then the driving gas enters the auxiliary drive cavity **213** to drive the auxiliary piston **31** moving upward relative to the main piston **21**; thus, the stem **41** is lifted together with the auxiliary piston **31**, causing the auxiliary disc **42** to rise and no longer close the deflation buffering hole **513**. At this point in time, as shown in FIG. 6, the working gas in the inlet **12** of the body **11** enters the chamber **52** via the deflation buffering hole **513** and then is discharged from the deflation buffering duct **53** to reach the outlet **13**. With the deflation buffering duct **53** being of a smaller diameter than the inlet **12**, and the deflation buffering hole **513** being of a smaller diameter than the inlet **12**, only a small amount of the working gas flows from the inlet **12** to the outlet **13**.

[0032] As shown in FIG. 7, the aforesaid operation is followed by the process of exiting the buffered opening state to enter the fully opened state, after the auxiliary piston **31** has risen to reach a predetermined height, the controlled gas source **91** (shown in FIG. 5) provides the drive gas to the drive air duct **25** such that the drive gas enters the drive cavity **153** to drive the main piston **21** rising relative to the drive cylinder **15**. Thus, the auxiliary piston **31** rises together with the main piston **21** such that the stem **41** and the auxiliary disc **42** rise. Therefore, the auxiliary disc **42** abuts against the upper abutting portion **511** (shown in FIG. 6) to lift the main disc **51** and stop the main disc **51** from closing the inlet **12**, allowing the working gas to flow from the inlet **12** to the outlet

13.

[0033] As shown in FIG. 3, returning to the shut state entails the following: controllably causing the controlled gas source **91** and the auxiliary controlled gas source **92** to absorb the driving gas, whereas the main disc **51** and the auxiliary disc **42** are pushed downward under the resilient restoring force of the main spring **55** and the auxiliary spring **35**, causing the main piston **21** and the auxiliary piston **31** to descend.

[0034] As explained through the above description of operation, this embodiment is advantageous in that the opening process is preceded by deflation buffering and followed by a fully opening process to optimize the effect of the deflation buffering. Furthermore, the disclosure is advantageous by providing a structure of joint operation of the stem **41**, the auxiliary disc **42** and the main disc **51** to drive the auxiliary disc **42** and the main disc **51** separately to not only achieve structural simplicity and ease of assembly but also ensure that all the components of the two-way angle valve operate smoothly. By contrast, the aforesaid conventional two-way angle valve has a main stem and an auxiliary stem which have to be driven separately to the detriment of smooth operation of some components of the conventional two-way angle valve.

[0035] It is noteworthy that the upward-downward relation mentioned herein merely serves exemplary purposes but does not necessarily apply to any upward-downward relations shown in the accompanying drawings. In practice, the upward-downward relation mentioned herein may be interpreted as an upside-down orientation or a side-mounted orientation.

[0036] The disclosure is disclosed above by embodiments. The embodiments are illustrative of the disclosure but shall not be interpreted as restrictive of the scope of the claims of the disclosure. Thus, all simple variations or equivalent implementation of the aforesaid embodiments according to the claims and detailed description of the disclosure shall be deemed falling within the scope of the claims of the disclosure.

Claims

1. A two-way angle valve capable of buffering deflation, comprising: a body having an inlet disposed below and an outlet disposed laterally; a drive cylinder coupled to the body and disposed on the body; a main piston driven to be vertically movably disposed in the drive cylinder, wherein an auxiliary drive cavity is concavely disposed at the main piston and extends in a top-to-bottom direction; a drive air duct not only disposed at the drive cylinder so as to be in communication with an inside of the drive cylinder but also externally connected to a controlled gas source; an auxiliary piston driven to be vertically movably disposed in the auxiliary drive cavity; an auxiliary spring disposed in the drive cylinder, having two ends abutting against the drive cylinder and the auxiliary piston respectively, and generating a resilient restoring force under which the auxiliary piston is moved downward; a stem having a top end fixedly disposed at the auxiliary piston, extending downward by a predetermined length, and moving vertically together with the auxiliary piston to penetrate the main piston and the drive cylinder to thereby move into the body, wherein an auxiliary disc is disposed at a bottom end of the stem; an auxiliary drive air duct comprising an auxiliary channel penetrating the drive cylinder, a gap defined between the stem and the drive cylinder, and a stem air duct disposed in the stem, wherein the auxiliary channel, the gap, and the stem air duct are in communication with each other, the stem air duct being in communication with an inside of the auxiliary drive cavity, and the auxiliary channel being externally connected to an auxiliary controlled gas source; a main disc having a chamber and a deflation buffering duct, the main disc having an upper abutting portion and a lower abutting portion, the upper abutting portion being disposed above the chamber, the lower abutting portion being disposed below the chamber and having a deflation buffering hole, the stem being inserted into the main disc to allow the auxiliary disc to be vertically movably disposed in the chamber, the deflation buffering duct having an end being in communication with the chamber and another end being defined as an opening

formed on a lateral side of the main disc, thereby allowing the deflation buffering duct to be in communication with the outlet, and the deflation buffering hole being in communication with the inlet; and a main spring disposed in the body, fitted around the stem, having two ends abutting against the drive cylinder and the main disc respectively, and generating a resilient restoring force under which the main disc is moved downward, wherein the stem and the auxiliary disc move because of vertical movement of the auxiliary piston, wherein movement of the auxiliary piston and the main piston causes the stem to move between a lowest position and a highest position, wherein when the stem is at the lowest position, the auxiliary disc abuts against the lower abutting portion of the chamber to close the deflation buffering hole, and the main disc abuts against the body to close the inlet, disabling communication of the inlet and the outlet, wherein when the auxiliary disc has separated from the lower abutting portion but has not yet abutted against the upper abutting portion during an ascent of the stem from the lowest position, the outlet is in communication with the inlet through the deflation buffering duct, the chamber and the deflation buffering hole, wherein when the stem is at the highest position, the auxiliary disc abuts against the upper abutting portion of the chamber, and the main disc is lifted such that the inlet is not closed, enabling the communication of the inlet and the outlet.

2. The two-way angle valve capable of buffering deflation of claim 1, wherein the deflation buffering duct has a smaller diameter than the inlet.

3. The two-way angle valve capable of buffering deflation of claim 1, wherein the deflation buffering hole has a smaller diameter than the inlet.

4. The two-way angle valve capable of buffering deflation of claim 1, further comprising a resilient sealing pipe with two ends connected to the main disc and the body respectively to cover the stem and the main spring.

5. The two-way angle valve capable of buffering deflation of claim 1, wherein the drive cylinder comprises a lower block and an upper cylinder fitted together, with the body disposed at the lower block, allowing a drive cavity to be formed between the upper cylinder and the lower block and adapted to receive the main piston and the auxiliary spring, with the stem penetrating the lower block.

6. The two-way angle valve capable of buffering deflation of claim 5, wherein the drive air duct and the auxiliary channel are disposed at the lower block.

7. The two-way angle valve capable of buffering deflation of claim 6, wherein the lower cylinder has an upper seal ring and a lower seal ring and is slidably fitted to the stem so as to allow a position of communication of the auxiliary channel and the gap to stay between the upper seal ring and the lower seal ring and allow a position of communication of the stem air duct and the gap to stay between the upper seal ring and the lower seal ring.
